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Dear Editor,

We would like to submit our manuscript "Pair Attribution of the VV10 Nonlocal-Correlation Term and Its Grid-Partition Dependence" as an Article in JCTC.

The question behind it is a narrow one. When a VV10 calculation gives you a number for a dimer, how much of that number belongs to the interaction between the two fragments? Production kernels cannot answer it. They fold the double sum over grid-point pairs into per-point values before returning, so the pair index is already gone. We looked for a published implementation that keeps it and did not find one.

So we kept the index and summed the pairs explicitly, assigning each grid point to a fragment through a real-space partition. On the eight S22 complexes at ωB97X-V/6-31G*, the inter-fragment block accounts for 0.92-1.08 of the frozen-density superposition difference. Almost all of the nonlocal correlation that changes on association sits between the fragments, not inside them.

The part we think matters more is what happens when you change the partition. The four Becke-family schemes disagree by 0.004-0.247 kcal/mol, which is 0.8-18.2% of the Becke value, and the disagreement does not order by interaction type. Two orientations of the same benzene dimer differ by a factor of eleven in the fragment-resolved percentage. Their total interaction energies also rank one way before counterpoise correction and the other way after it. A single fragment-resolved dispersion number is therefore not a well-defined quantity, and we would rather say so than publish one scheme's value as if it were. The paper reports the range and gives a way to estimate it per system.

We also kept a negative result in the main text. We built a certified screening scheme for the pair sum and timed it against the exact sum. None of the 33 screened/exact pairs, measured in two independent windows, ran faster. The certificate available at atom-pair granularity is loose by about 2.4e5 relative to what it would have to decide, so the traversal cannot recover the cost. Readers planning a similar optimization should know this before they spend time on it.

On verification: ORCA 6.0.1 reproduces our nonlocal differences to under 1% of the term on all eight systems, binding energies are checked against CCSD(T)/6-31G*, and every number in the paper is mapped to an archived output in the repository. The code, nlcsplit, is Apache-2.0 and public at https://github.com/Haangyeon/nlcsplit. It contains no GAMESS source.

Thank you for considering it.

Sincerely,

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